ERRAMANZIL ZILLA PARISHAT HIGH SCHOOL PROJECT, HYDERABAD, TELANGANA, INDIA

Executed By

EWB India, EWB MJCET Student Chapter and Nirmaan

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1. EXECUTIVE SUMMARY

EWB India along with EWB MJCET Student chapter with support from Nirmaan - an NGO, conducted extensive needs assessment survey at Erramanzil High School and identified infrastructure and spoken English and computer education as the required interventions to improve the overall quality of education at this school. Erramanzil Zilla Parishat High School, a government of Telengana School, is located in Erraamanzil, Hyderabad, Telengana, India.The objective of this project is to improve the intake of students and also impart holistic and quality education to the students.

The assessment identified that the school needs improved drainage system, hygienic wash rooms for male and female students and staff, clean drinking water, decent playground for students, sports kits, white wash of school building, better black boards and proper lighting in class rooms. It was also noticed that the students need support in spoken English and computer education.

MJCET EWB Student chapter under the guidance of EWB India spear headed and implemented the infrastructure aspect of the project. They developed the infrastructure project plan and estimated budget and submitted to EWB India for review and consideration. MJCET EWB Student chapter implemented the identified infrastructure requirements in a span of three months and the total budgeted cost was around USD 11,000 (@ an exchange rate of 1USD = 65 INR).

We would like to offer our gratitude and thanks to Bechtel Corporation in USA who supported us providing the funding through EWB International and Ms Cathy Leslie Executive Director Ewb USA to make it happen.

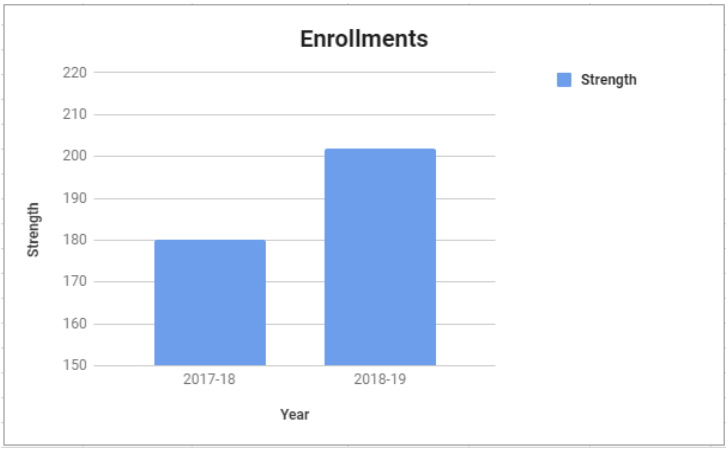
EWB India along with Nirmaan embarked on improving the overall quality of education. EWB India joined hands with Nirman, who is the knowledge partner in improving quality of education in government funded schools. Nirmaan is a registered NGO formed by students of BITS Pilani in 2005 and works in the areas of education, livelihoods and social leadership. Nirmaan touched more than 3,00,000 beneficiaries through vertical flagship programs and around 5,00,000 beneficiaries through horizontal impact programs with 150 full time employees and 500 strong volunteer network, with 500 Lakhs (PA) worth of social impact through 7 flagship programs & social leadership initiatives touching lives of children, women, youth and

farmers from diverse social backgrounds across 7 states of India partnering with 35 corporate and philanthropic partners.Nirmaan conducted detailed survey on the health of the education at the school and came up with a road map to increase the student intake and also to impart holistic education system. Accordingly, two teachers were hired to teach spoken English and computers for a period of two years. EWB India has communicated a budget of approximate 2000$ towards teh salaries for the English and computer teachers for an academic year.

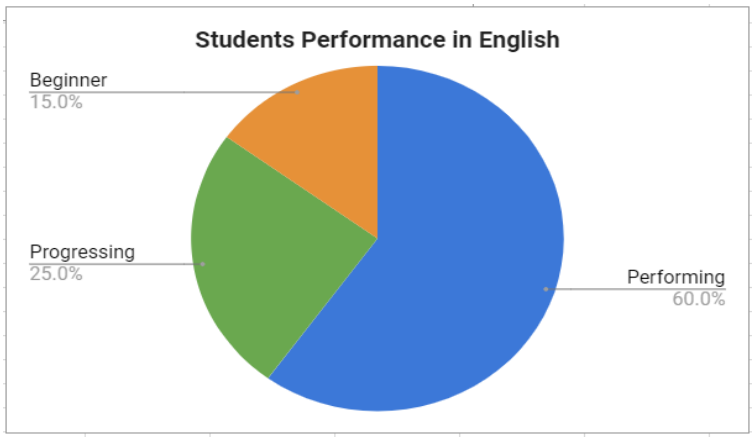
EWB India observed that there has been an increase in student intake and significant improvement in the spoken English abilities of the students and computer education levels in a short duration of 6 months.

The following is the summary of the impact of this project on the overall quality of education at the school.

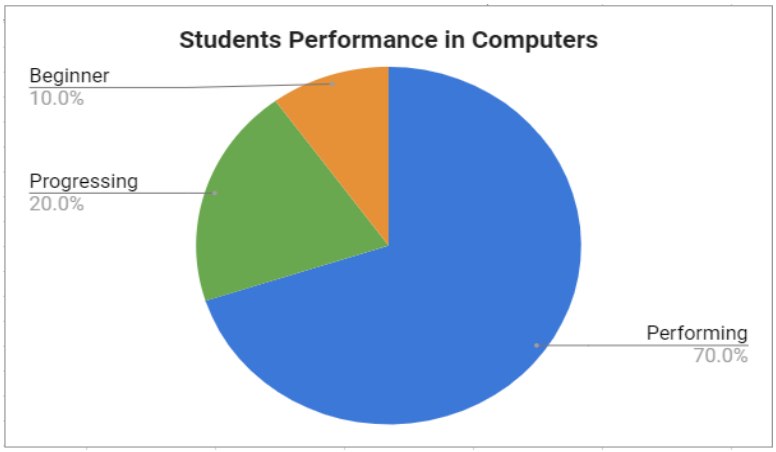
Increase in Enrolment: The student strength has increased from 180 in the year 2017 to 202 in year 2018



Improvement in communication in English Language: 60% of the students are able to introduce themselves in English and able to write general essays, 25% of the students are in progressing stage and 15% are very slow learners.

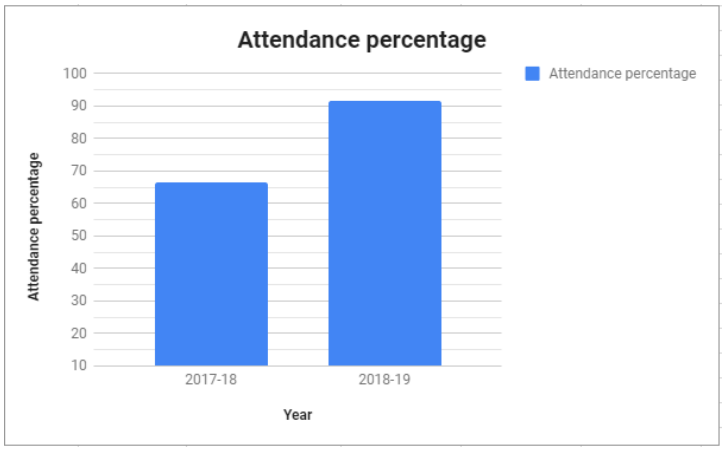


Improvement in Computer Education: 70% of the students are able to work on computers and draw their own paintings, 20% are in progressing stage and 10% are regular absentees.



Increase in student attendance: Now the students are very enthusiastic and eager to come to the school because of computers and English digital classes hence attendance of students increased. Absenteeism has decreased and regular strength has increased from 120 to 185

Year	Total Strength	Regular Strength	Attendance
2017-18	180	120	66.60%
2018-19	202	185	91.50%



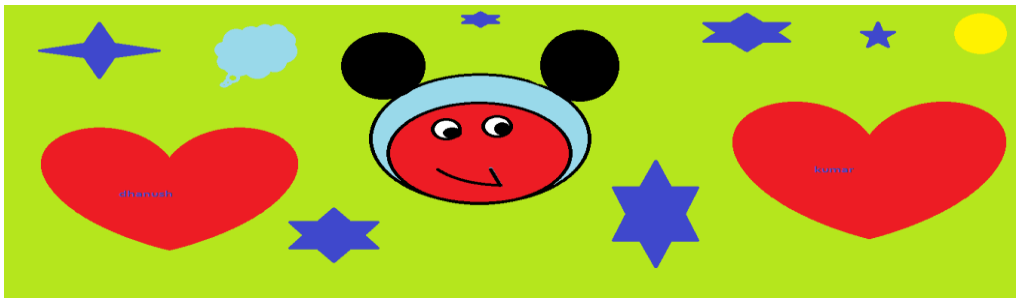
Paintings by the Students:



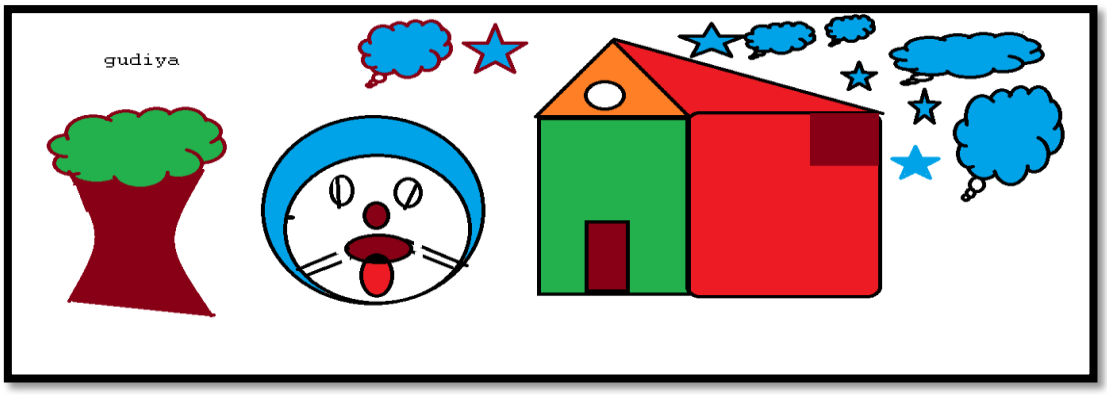
John, 9th Class Student



Shivaram, 8th Class Student



Dhanush, 7th Class Student



Gudiya, 7th Grade Student



Satish and Sai, 6th Grade Students

EWB India team will continue to monitor the progress on a continued basis for improving the quality of education.

EWB India would like to thank EWB MJCET Student chapter and Nirmaan for their unrelenting support in developing the school infrastructure and improving quality of education. There are also other stakeholders who supported the vision of EWB India and EWB India would like to thank all the stakeholders and we look forward to their continued association and support. EWB India would like to undertake the second phase of the project by focusing on developing the required infrastructure and improving the quality of education at the primary school.

Future Proposal

It is proposed to extend the spokenEnglish and computers teaching support for an additional one year Budget needed – $4000/-

There is a primary school in the same school complex requiring infrastructure improvement.

Preliminary Assessment indentified the needs to be providing clean drinking water , improve sanitation, overall school painting and providing black boards, chairs and hall with shelter facilities. Budget need - $10,000/-

PROJECT SNAPSHOT

Project Name	Erramanzil High School Infrastructure Development Project
Location	Erramanzil,Khairtabad,Hyderabad,Telangana State,India
Assessment Observations	Poor drinking water facilities, improper sewage system, out-datedcomputer systems, lack of school playground, lack of washing area, electricity, peeling of plastering, painting, water leakage, lack of teachers to teach computers and spoken Englishetc . As a result of these issues, students were showing limited interest in attending the school and continuing their education.

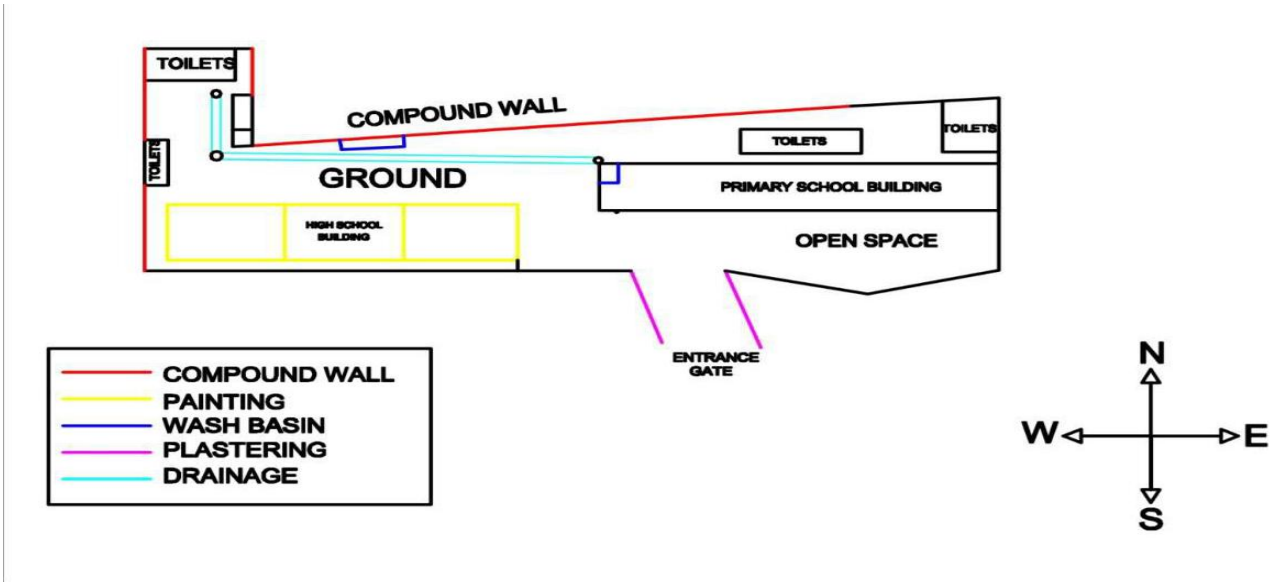
Project Objectives	The objective of this project was to develop the infrastructure in school premises, increase the student's intake, enhance the academic performance of the students, and improve the quality of education imparted. The focus was to create a conducive atmosphere where by students voluntarily attend the school and enjoy the education process.
Stake holders	EWB MJCET Student chapter, EWB India and Nirmaan. EWB MJCET Student chapter implemented the infrastructure projects under the guidance and support from EWB India. Nirmaan is the knowledge partner who is sharing its experience in improving the quality of education. All the three stakeholders are periodically monitoring the overall performance of the students and intervening accordingly.
Impact	The student intake has improved from 180 in the year 2017 to 212 in the year2018.Students and staff have now access to clean and safe drinking water, improved wash rooms, better playground, better E-learning centre. EWB MJCET Studentswere also involved in plantation of saplings for healthier & greenerenvironment. Spoken English teacher and Computer teachers are continuously monitoring the performance of the students. It was noted the spoken English has improved and the awareness and usage of computers has increased due to the interventions.

2. Project Background

The infrastructure and overall conditions of Government High School, Erramanzil was not up to the required standard. So, we the students of Engineers without Borders-MJCET, in collaboration with EWB-India and Nirmaan foundation adopted Government high school, Erramanzil, Hyderabad, Telangana, India for developing the school completely in terms of infrastructure, providing better facilities to the occupants,providing career guidance andshowing the importance of hygiene for an overall better quality of education.

2.1 Government. High School, Erramanzil, Hyderabad

It consists of 190 students with overall staff (Teaching and Non-Teaching) comprising of 30 members working under government of Telangana established in late 1990's.



A 2-D representation of the School Campus



3. Activities undertaken by EWB MJCET Student Chapter:

EWB MJCET Students were divided into several teams and developed the following:

1. RO Plant
2.Drainage system
3.Developed playground
4.Installed hand wash platform
5. Planted trees
6.Implemented drip Irrigation
7.Constructed broken compound wall
8.Repaired doors
9.Implemented plumbing works
10.Welded gates and doors
11. Computer software development
12. Plastering of walls
13. Painting of black boards
14. Conducted Career counselling sessions for students

3.1 R O Plant:

Clean water was one of the most neededrequirements of the school.

The existing RO plant was not functioning. The team started with the repair of RO plant by replacing the mother board panel which was not working. The sand and carbon which helps in the purification of waterwas decayed, new sand and carbon wasreinstated in the plant. New candles were also put into place.

The plant was thoroughly cleaned by a chemical wash i.e. with H2 and O2; the foul particles were cleaned from membrane of theRO plant.

After the cleaning of plant, the water which was going to be utilised was tested for purity with a hardness test.

Finally, the team made a supporting platform for the RO plant, a dedicated facility for accessing of clean drinking water.

Before: RO Plant



After: RO Plant and Drinking Water Platform:



3.2 Drainage system:

A major problem the school had faced was an open drainage system which could have been the cause for various harmful diseases if not attended. Drainage pipelines were either in a dilapidated or a clogged condition leading the drainage waste to overflow from the man holes. Unclogging the pipes wasn't the solution as the pipes that were in use were completely damaged or distorted.

Team members from civil engineering provided a layout for placing of new pipelines and manholes system.With the help of excavators and labours new drainage system was constructed and checked for proper functioning of drainage system.

Before:Old Drainage System





After: New Drainage System





3.3 Play Ground Development:

As the famous saying goes “All study and no play makes akshita a dull girl”. Physical activity is a vital part of childhood that helps children develop physical strength, coordination and balance.

The playground in this school was uneven and water-clogged which was not only uncomfortable for the students to play but also poses a potential harm while playing. The team decided to take up the task of creating a safer play zone.

Ground surface was made even by levelling of the playground. An excavator was put into action to remove all the unwanted stones and other materials which were present on ground area and the surface was made even. After the levelling, the playground was coated with red sand and a roller was utilised to spread out the sand uniformly throughout the playground for a smoother playground.

Before: Playground



After: Improved Playground



3.4 Hand Wash Platform:

The hand wash platform that was in use was in a very poor condition and it was necessary to replace it with a new one.

New and improved hygienic hand wash platform for the school was constructed.

Before:



After: Clean and Hygienic Hand Wash System



3.5 Tree Plantation:

There is a saying **"**healthy mind lives within healthy body". We all know fresh air not only gives us healthy lungs but also produces healthy minds which are essential for children in a learning environment.

So, after mapping out strategic points throughout the school campus, students planted several trees and shrubs including mango and neem trees etc.

These saplings were acquired from various government nurseries and students planted saplings and watered them.





3.6 Drip Irrigation:

Drip irrigation is type of micro-irrigation of soil and has the potential to save water and nutrients by allowing water to drip slowly to the roots of plants, either from above the soil surface or buried below the surface.

An arrangement was provided for storing of excess water coming out from R O plant in storage tank and connections were provided to allow the flow of this stored water into soil through an underground pipeline system which can be operated with a valve.

3.7 Compound Wall Construction:

Compound wall construction was a necessary needas it providessafety for the school premises and also it prevents unwanted intruders and the dumping of garbage from the surroundings.So our team constructed a compound wall of height 6ft and of 9inch thickness around the school building where there was no wall constructed.

Before: Compound Wall



After: Compound Wall



3.8Door Repairing:

Some of the doors in the school premises were completely damaged that were replaced with new ones by students under the guidance of professionals.

Before: Doors



After: Doors



3.9Plumbing &Fittings

One of the major problems the school was facing was poorly maintained bathrooms with bad plumbing. The sanitation team surveyed the situation and came up with the best possible solutions. Students connected pipelines from over-head tank to the internal plumbing in the bathrooms. The distance from over-head tank to toilets was quite long.

According to the required measurements, the PVC pipes were purchased along with the hardware materials required for the purpose. A professional was also consulted for suggestions and improvements in the design.

Some of the pipeline work was done by the students using their engineering skills.



3.10Welding :

Welding was done at various required points ondoors, windows and main gate of the school building.

This welding work was done by mechanical engineering students utilising their engineering skills under the guidance of professionals.



3.11Computer Software Installations:

School computer lab was fully equipped with computers but the operating system in place was not user friendly and suitable for educational purpose.

TheTech. team updated the computers which is user friendly for the students. Several E-Learning guides and materials was installed which will be beneficial for the students.





3.12Plastering of wall:

Many walls with cracks and holes were plastered uniformly.

Before: Walls



After: Walls



3.13Painting:

Painting helps in the protection of walls from badweather; also it improves the aesthetics of the school building and contributes towards the boosting of morale for the teachers and students.

All the surfaces of wall were plastered and wall treatment was done to protect them from rain and scorching heat.

Before:



After:



3.14Wash Area:

This area was built for washing and general purpose for the students.



3.15Career Counselling Session:

This session was carried out for developing and planning of student's career. This was done by 20 volunteers of EWB MJCET Students and in collaboration with Nirmaan Foundation. In this session studentswere provided with various career paths that can be adopted after completion of their 10th grade.





4. Recommendations:

- It was suggestedthatproper maintenance schedule be put in placefor the newly installed RO Plant and hand wash platforms.
- Periodicsurveysmust be undertaken to assessthe academic performance of the students.
- Students must be encouraged to use the E-Learning facilities that are available in the school.

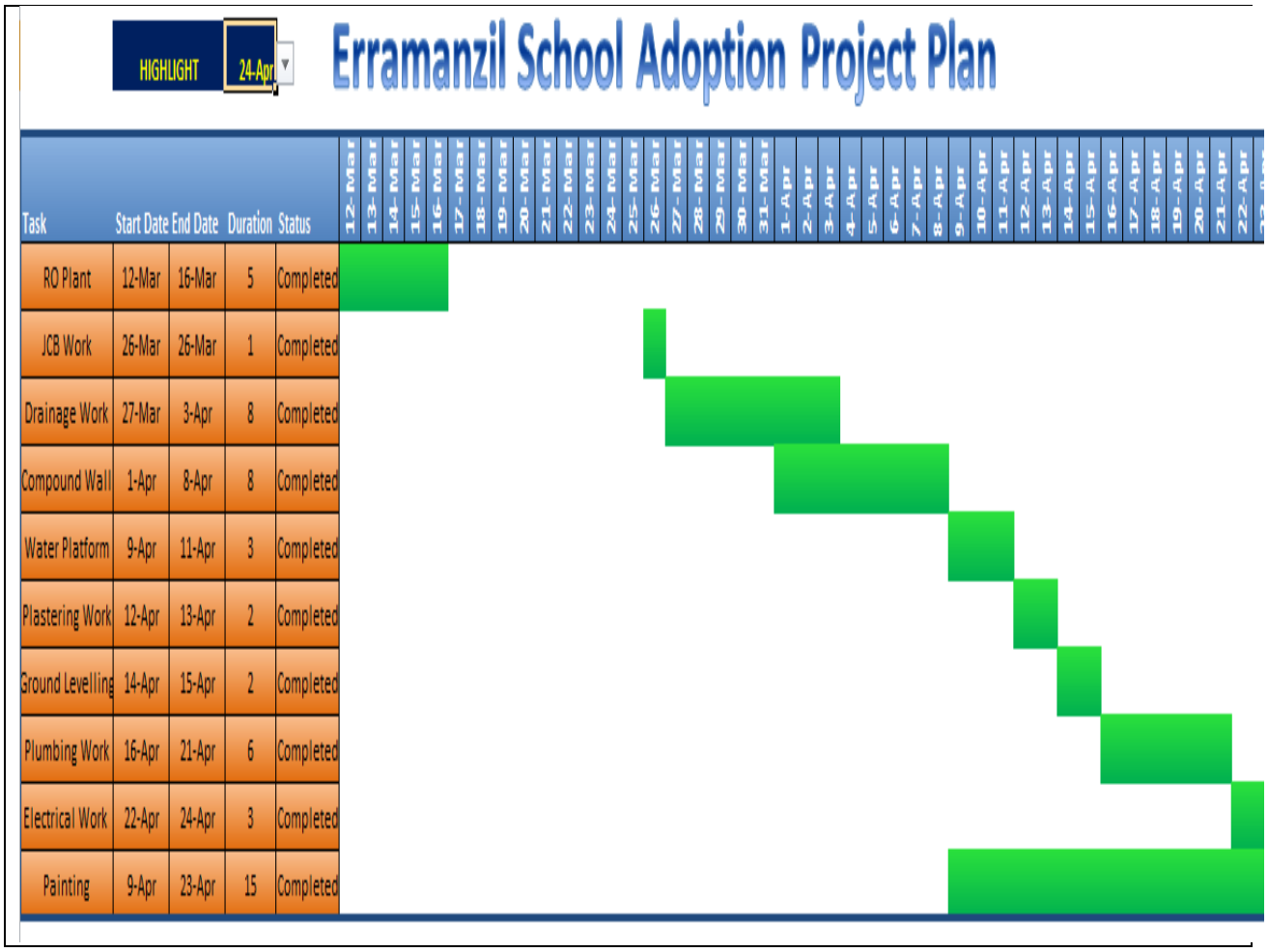
5. Future Proposed Activities:

Infrastructure activities were undertaken and implemented for the High School. Academic interventions for the High School are in progress. The primary school students needs infrastructure support such as furniture, painting, tubes, fans, clean drinking facility, wash rooms, etc. It is proposed to undertake these activities under Phase-II of the EWB India Erramanzil High School development activities.

ANNEXURE-I

The following is the project plan which was adopted for the execution of the project:

GANTT CHART



ANNEXURE-II

(COST OF SCHOOL INFRASTRUCTURE DEVELOPMENT)

S.No	Area of work	Amount Expended	Amount in $
1.	R O Plant	Rs 15,500/-	$240/-
2.	Drainage system	Rs 1,08,940/-	$1680/-
3.	Compound Wall& other plastering works	Rs 1,39,380/-	$2200/-
4.	Playground development	Rs 58,450/-	$900/-
5.	Carpentry Works	Rs 33,720/-	$520/-
6.	Drip Irrigation	Rs 7,870/-	$120/-
7.	Plantation	Rs 3,800/-	$60/-
8.	Welding Works	Rs 7,900/-	$120/-
9.	Plumbing Work	Rs 26,660/-	$400/-
10.	Construction of Wash Area	Rs 16,750	$300/-
11.	Painting	Rs 2,30,000/-	$3600/-
12.	Construction of Hand Wash Platform	Rs 32,800/-	$500/-
13.	Drinking Water platform	Rs 13,710/-	$210/-
14.	Petty works	Rs 19,540/-	$300/-
	Total Cost	Rs 7,15,020/-	$11150/-

ANNEXURE III

News and press coverage of the School Development project



సాక్షి హైదరాబాద్ 6

ఖైరతాబాద్

పోలీస్ అధికారులకు

మరిన్ని స్కూల్స్‌ను దత్తత తీసుకుంటాం

ఎర్రమంజిల్ ప్రభుత్వ పాఠశాలను దత్తత తీసుకుంటామని ప్రకటిస్తున్న ఉస్మానియా కాలేజ్ విద్యార్థులు

ఆవరణలో మొక్కలు నాటుతున్న ఉస్మానియా కాలేజ్ విద్యార్థులు

- ఎర్రమంజిల్ ప్రభుత్వ పాఠశాలను దత్తత తీసుకుంటామని ప్రకటిస్తున్న ఉస్మానియా ఇంజినీరింగ్ కాలేజ్ విద్యార్థులు
- స్కూల్‌లో మౌలిక సదుపాయాలు ఏర్పాటు
- భవనానికి పూర్తి పెయింటింగ్
- ఆవరణలో మొక్కలు నాటిన దండయ్యాదీ సభ్యులు

స్కూల్‌ను దత్తత తీసుకున్నాం

అధ్వర్యంగా భావిస్తున్నాం

సాధ్యమైనన్ని పనులు మేమే చేస్తాం

మీ ప్రాంత సమస్యలపై సమాచారమివ్వండి..

బంజారాహిల్స్	95055 07523
పంజగుట్ట	98482 74231
ప్రకాశ్‌నగర్	96525 66828
సోమాజిగూడ	94945 53555
సికింద్రాబాద్	9505590393

ANNEXURE IV

Overview of the Project by EWB India MJCET Student Chapter Team Leader:

"At EWB-MJCET we are nurtured to work for the betterment for society. We had a team of 44 members who contributed in the successful accomplishment of the project. I thank EWB-India and Dr. Ashok Agarwal sir for providing us with an opportunity to give something back to the society. It was a great experience working for the development of the school which will aid the students in pursuing a better quality of education."

Mohammed Sohel Khan Imran

President, EWB-MJCET

Feedback from the Principal

Computer and English communication sessions are very useful to the students. Students are able to introduce themselves and they are able to tell about their school also. Communication skills of the students increased. 9th and 10th students are showing interest to speak in English.

We have fully equipped computer lab but as there is no trainer, we are not at all using the lab. After appointing the computer trainer, computer practical sessions are conducted to the students. These sessions are also useful for their other subject project works. Thank you so much to EWB India, MJCET and Nirmaan for the support.

Shanta Rathod, Principal

Feedback from the English Teacher:

Here, the students are very eager to learn new skills. There are students who are quick learners as well as slow learners. For this reason, I am teaching them in an innovative manner, using Smart-Class, where students are learning by visuals. Even students are more interested in Digital Class than regular classes.

Varsha, English Teacher

Feedback from the Computer Teacher:

Most students are learning, and they are learning fast. Students have improved a lot in terms of using a computer and their practical knowledge has improved and I am confident that I will be able to finish all the topics within the given time.Improvement is needed in terms of being able to communicate what they have learned which they will be able to do by the end of the course.

Omkar, Computer Teacher

Feedback from EWB India:

The experience of working with government High school, Erramanzil along with Nirmaan and our EWB MJCET Student chapter has been really rewarding. Our involvement illustrated that quality of education can be significantly imposed by pounding a proper echo system with good infrastructure and good teachers. We look forward to the continued involvement in this school and hope to make a difference.

Dr Ashok Agarwal
Director, EWB – India.

Feedback from Nirmaan Organization:

EWB India MJCET Chapter done a great work in Erramanzil. Students from MJCET worked very hard to deliver the project, entire school atmosphere is changed because of their work at the school. We, Nirmaan is very excited to work with them. Trainers who appointed in the school are doing good, with their sessions there is a change in academics of the students as well. Thank you for choosing Nirmaan as knowledge partner for the project. We are looking forward for the more opportunities to work with EWB.

Mayur Patnala,
President, Chief functionary,
Nirmaan Organisation.

Annexure V

Project Pictures:







